GUILHERME HENRIQUE RISSON SILVA

Software Engineer | Backend | Python | FastAPI | PostgreSQL
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SUMMARY

Early-career Software Engineer and Software Engineering student at Uninter. At Vitru Educação I build the Python systems behind UniCesumar's academic planning, including an instructor allocation optimizer running in production (FastAPI, React, TypeScript, OR-Tools). In my own projects I build and ship backends with FastAPI, PostgreSQL and LLMs, with automated tests, CI on GitHub Actions and architecture decisions recorded as ADRs.

EXPERIENCE

Vitru Educação (UniCesumar) — Hybrid, Maringá, Brazil
Mid-level Data Analyst, Academic Planning — 03/2026 - Present

- Built and maintain in production the instructor allocation optimizer: FastAPI API, React and TypeScript interface and OR-Tools CP-SAT solver. Assigns about 200 instructors to more than 350 classes per module in minutes, work that used to take weeks in spreadsheets.
- Maintain in production the weekly curriculum pipeline for UniCesumar and Uniasselvi (Python, SQL Server), with 6 input layouts, 4 validation layers and an auditable history.
- Moved the planned-vs-actual audits from Power Query to Python, over datasets from 1 million to more than 40 million rows.
- Built the 2027 student enrollment forecasting model (Python, DuckDB, Parquet), with 50 automated quality checks and error below 2% against actuals.
- Rewrote in Python and Marimo the lab class demand vs. tutor capacity model, with a simulator by campus and city and Power BI dashboards for managers and leadership.

Linoforte Móveis — Osvaldo Cruz, Brazil
Assistant Operator — 06/2022 - 12/2025

- Set up monitoring of production and quality KPIs, flagging recurring stoppages for corrective action.
- Mapped bottlenecks and downtime and proposed improvements that reduced rework and waste on the line.

PROJECTS (github.com/guilhermerissondev)

- **Job Hunters**: live job-matching SaaS (job-hunters-ebon.vercel.app) with FastAPI, SQLAlchemy, Alembic, PostgreSQL/pgvector and Next.js. Collects about 98k jobs from 649 sources, matches them with evidence from the CV and delivers them on Telegram. Deployed on Vercel, Render, Neon and Supabase, CI on GitHub Actions, 380 tests and 21 ADRs.
- **Teacher Allocation Optimizer**: the optimizer running in production at Vitru, open source (FastAPI, SQLite, React, TypeScript, OR-Tools CP-SAT), with a scenario engine isolated from the main solver.
- **TikTok Viral Generator**: agent that researches, scripts and publishes 4 short videos a day with a source for every fact (Python, Pydantic, SQLite, ffmpeg). 691 automated tests.
- **Regulatory RAG Eval**: RAG over 294 Brazilian Central Bank regulations with source citations (BM25, embeddings, RRF). Hybrid search raised MRR from 0.361 to 0.644.
- **Procurement Triage Service**: FastAPI service that extracts data from Brazilian public procurement notices (PNCP) with an LLM and only accepts values found in the PDF. 61 offline tests in CI, US$0.028 per notice.

EDUCATION

B.S. in Software Engineering - Uninter - 10/2026 - 2030 (in progress)
AI Software Engineer 4.0 Program (360 h) - Data Science Academy - 09/2026 - 12/2026 (expected)
Postgraduate Certificate in Data Science (360 h) - UniCesumar - 03/2026 - 10/2026
B.S. in Civil Engineering - FAI, Faculdades Adamantinenses Integradas - 2021 - 2025

SKILLS

Languages: Python, TypeScript, SQL
Backend: FastAPI, Pydantic, SQLAlchemy, Alembic, REST APIs, pytest, Docker, GitHub Actions (CI/CD), Git, Linux
Frontend: React, Next.js, Tailwind CSS
Databases: PostgreSQL, pgvector, SQL Server, SQLite, DuckDB
AI and LLM: RAG, hybrid search (BM25 + embeddings), structured output, LLM evaluation, agents, Gemini API
Data: pandas, Parquet, OR-Tools (CP-SAT), Power BI
Spoken languages: Portuguese (native), English (advanced), Spanish (basic to intermediate)